

PINN_M3DC1

Module 3

Dataset Contracts and M3D-C1-Shaped Input/Output

A textbook-style scientific data specification for auditable reduced-MHD surrogate learning

Module purpose

Store accepted fields, parameters, coordinates, provenance, whole-case splits, and normalization in explicit versioned schemas while preserving the physical meaning fixed by Modules 1 and 2.

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Physics and data-contract note for the synthetic reduced-MHD proof of concept

Preface: why a data contract is part of the physics

Module 2 produces verified numerical trajectories. Module 3 determines whether those trajectories remain scientifically intelligible after they are written to disk, combined into a dataset, split for learning, transformed for optimization, read by another programming language, and eventually compared with genuine M3D-C1 exports.

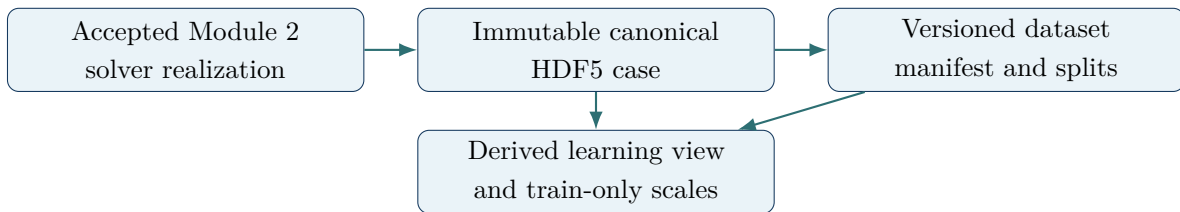
A file format alone cannot answer these questions. An array of numbers does not identify which axis is time, whether current means $J = -\nabla_{\perp}^2 \psi$ or $J = +\nabla_{\perp}^2 \psi$, whether the rightmost spatial endpoint is duplicated, whether a field is dimensional or nondimensional, whether a case passed convergence checks, or whether preprocessing used the test set. Those facts determine the scientific meaning of every number. The **data contract** is the set of rules that makes that meaning explicit and machine-checkable.

The initials **I/O** mean *input/output*: how information enters and leaves a computational component. **HDF5** means Hierarchical Data Format version 5, a format that organizes multidimensional datasets, groups, and attributes. **Schema** means a versioned specification of required names, types, shapes, meanings, and relationships. **Provenance** means the recorded history of how an artifact was produced. **Normalization** here can mean either physical nondimensionalization or a later numerical transform for neural-network training; these are different operations and must remain distinguishable.

Important distinction

“M3D-C1-shaped” describes the architecture of the workflow: coordinates, parameters, time-dependent fields, metadata, diagnostics, and an adapter boundary. It does not mean that the synthetic Cartesian variables are numerically equivalent to M3D-C1 variables, that the geometries agree, or that the dataset validates an M3D-C1 surrogate.

The main dependency chain



The reference case is never overwritten by a preprocessing step. A learning-ready representation is a derived artifact whose parent, transforms, split manifest, and checksums are recorded.

Learning objectives

After working through this module, the reader should be able to:

1. explain why array names and shapes are physical statements rather than clerical details;
2. distinguish a physical case, a numerical realization, a snapshot, a grid point, and a supervised example;
3. state the exact canonical shapes and meanings of ψ , U , J , and ω ;
4. explain why the stored reference fields remain unstandardized and immutable;
5. design an HDF5 hierarchy that preserves coordinates, fields, parameters, diagnostics, verification, and provenance;
6. distinguish required datasets from metadata attributes;
7. assign stable identifiers without encoding ambiguous floating-point strings in filenames;
8. construct whole-case training, validation, interpolation-test, and extrapolation-test partitions;
9. identify leakage caused by pointwise splitting, duplicate realizations, or test-fitted normalization;
10. fit and store invertible preprocessing transforms using training cases only;
11. flatten field arrays for minibatching without losing reconstruction indices or case balance;
12. verify Python–MATLAB axis, dtype, string, and checksum parity;
13. decide when a schema change is backward compatible and when it changes scientific meaning;
14. describe the explicit translation work required for genuine M3D-C1 data; and
15. state precisely which claims a Module 3 dataset can and cannot support.

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1 Scientific role and acceptance boundary

1.1 The object produced by Module 3

For a physical parameter vector

$$\boldsymbol{\mu}_c = (\eta_c, \nu_c, A_{0,c}), \quad (1.1)$$

Module 2 approximates a reduced-MHD trajectory. Module 3 creates an auditable dataset

$$\mathcal{D} = (\mathcal{C}, \mathcal{M}, \mathcal{S}, \mathcal{T}), \quad (1.2)$$

where

- $\mathcal{C}$ is a set of accepted canonical case files;
- $\mathcal{M}$ is a manifest describing and indexing those files;
- $\mathcal{S}$ is an immutable whole-case split assignment; and
- $\mathcal{T}$ is the set of declared, invertible preprocessing transforms.

The manifest is part of the scientific artifact. A directory of HDF5 files without an authoritative index does not define which files belong to a release, which are accepted, or how they are partitioned.

1.2 Four layers that must not be confused

Layer	Object	Main question
Physical model	Continuous fields and equations	What do $\psi, U, J, \omega, \eta, \nu, A_0$ mean?
Numerical realization	Finite arrays from one solver run	Which grid, time step, integrator, precision, and verification status produced them?
Serialized reference	HDF5 datasets plus metadata	Can another reader reconstruct the same logical arrays and meanings?
Learning view	Selected or transformed examples	Which split, sampling rule, scales, and inverse transforms define the neural task?

A neural preprocessing choice must not be written back into the physical-model layer. Similarly, changing a field sign cannot be treated as a harmless file-format revision.

1.3 Acceptance inherited from Module 2

Module 3 ingests only cases for which the Module 2 acceptance record is complete and true. At minimum, the case must have passed:

- operator and sign tests;
- analytic or manufactured-solution checks;
- temporal and spatial convergence checks;
- spectral and current-sheet resolution checks;
- energy, periodicity, gauge, and topology checks;
- finite-value and complete-horizon checks; and
- Python–MATLAB parity at the declared level when parity is required for that release.

Validation checkpoint

The Module 3 validator must fail closed. Missing acceptance metadata means “not eligible,” not “probably accepted.” A rejected or incomplete case may be retained for debugging, but it cannot appear in an accepted release manifest.

1.4 What Module 3 does not establish

Module 3 does not prove that the solver equations are physically adequate for a stellarator, that a neural network has learned them, or that an M3D-C1 export can be interpreted through the same variables. It establishes a trustworthy interface for the stated synthetic model and a place where a future adapter can make every required translation explicit.

2 Frozen physical semantics

2.1 Domain, coordinates, and endpoint convention

The version-1 domain is

$$\Omega = [0, L_x) \times [0, L_y), \quad L_x = L_y = 2\pi. \quad (2.1)$$

The spatial grids are

$$x_i = i\Delta x, \quad \Delta x = L_x/N_x, \quad i = 0, \dots, N_x - 1, \quad (2.2)$$

$$y_j = j\Delta y, \quad \Delta y = L_y/N_y, \quad j = 0, \dots, N_y - 1. \quad (2.3)$$

The right endpoints L_x and L_y are not duplicated because they represent the same periodic points as zero. Saved times are an explicit array

$$t_n, \quad n = 0, \dots, N_t - 1, \quad (2.4)$$

and need not be reconstructed from a presumed constant save interval.

2.2 Fields and signs

The physical meanings frozen in Module 1 are

$$\mathbf{B}_\perp = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x), \quad (2.5)$$

$$\mathbf{v}_\perp = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x), \quad (2.6)$$

$$J = -\nabla_\perp^2 \psi, \quad (2.7)$$

$$\omega = +\nabla_\perp^2 U. \quad (2.8)$$

The evolution equations are

$$\partial_t \psi + [U, \psi] = \eta \nabla_\perp^2 \psi, \quad (2.9)$$

$$\partial_t \omega + [U, \omega] = [J, \psi] + \nu \nabla_\perp^2 \omega, \quad (2.10)$$

with bracket

$$[f, g] = f_x g_y - f_y g_x. \quad (2.11)$$

The canonical stored field names are case sensitive:

$$\text{psi}, \quad \text{U}, \quad \text{J}, \quad \text{omega}. \quad (2.12)$$

Aliases such as `u`, `phi`, `current`, or `w` are not accepted at the canonical boundary unless an adapter maps them and records the mapping.

2.3 Primary, prognostic, reconstructed, and derived

The labels “primary” and “derived” depend on purpose.

- The Module 2 time integrator advances ψ and ω .
- It reconstructs U from $\nabla_\perp^2 U = \omega$ with zero spatial mean.
- It derives $J = -\nabla_\perp^2 \psi$.
- The Module 4 and Module 5 neural networks use (ψ, U) as primary targets and derive (J, ω) from spatial derivatives of their predictions.

All four reference fields are stored because they support round-trip verification, derivative-sensitive scoring, and auditing. Metadata must still record the dependency identities in Equation (2.8); four stored arrays are not four independent physical degrees of freedom.

2.4 Gauge conditions and means

The stream function has an arbitrary spatial constant because only its derivatives enter velocity. The frozen gauge is

$$\langle U \rangle(t) = \frac{1}{L_x L_y} \int_{\Omega} U \, dA = 0 \quad \text{for every saved time.} \quad (2.13)$$

The spatial mean of ψ is recorded and must remain consistent within a case. Validators check the numerical means rather than trusting a metadata declaration.

2.5 Physical nondimensionalization

All version-1 case values are dimensionless under `dimensionless_v1`. The metadata nevertheless stores the normalization name and reference scales. A value of unity is still a scale choice; omitting it makes a later dimensional reconstruction impossible to audit.

Important distinction : stored truth versus learning values

The canonical fields are stored in the physical nondimensional variables above. They are not mean-centered, standardized, logarithmically transformed, clipped, or cast to single precision for neural training. Those operations belong to a derived learning view.

3 Identity and dataset hierarchy

3.1 Case, realization, snapshot, point, and example

Definition

A **physical case** is an initial-value problem with fixed equations, domain, boundary conditions, initial-condition formula, parameters, and intended time horizon. A **numerical realization** is one discretized solution of that physical case with a specified solver, grid, time step, precision, and source revision.

One physical case can have several realizations: coarse and fine grids, Python and MATLAB implementations, or repeated verification runs. A snapshot is one full spatial state at t_n . A grid point is one pair (x_i, y_j) . A supervised example is a selected input-target pair constructed from one array element.

3.2 Why two identifiers are required

The contract uses:

- `physical_case_id`: invariant across scientifically equivalent numerical realizations; and
- `case_id`: unique to one canonical numerical realization.

The physical identifier is derived from canonical physics-defining metadata: equations version, conventions version, domain, boundary conditions, initial-condition name and version, parameter vector, and intended output horizon. The realization identifier additionally includes discretization, integrator, precision, implementation identity, and source/config fingerprints.

Important distinction : duplicate-realization leakage

Splitting by `case_id` alone can place a 128^2 realization in training and a 256^2 realization of the same physical case in test. That is not an unseen-parameter test. All realizations sharing a `physical_case_id` must be assigned to the same partition.

3.3 Opaque identifiers and readable metadata

Identifiers should be stable and opaque, for example

$$\text{syn-p-8f3a10c64e9b}, \quad \text{syn-r-54b3027976ad}. \quad (3.1)$$

Floating-point parameters do not belong only in a filename. Strings such as `eta0.001` can be formatted inconsistently, rounded, or confused with a different unit system. Human-readable parameters remain explicit datasets or manifest fields; the identifier provides stable identity.

3.4 Release hierarchy

The logical hierarchy is

$$\text{campaign} \supset \text{dataset release} \supset \text{physical cases} \supset \text{realizations} \supset \text{snapshots} \supset \text{grid points}. \quad (3.2)$$

A campaign records the parameter-study intent. A dataset release freezes a particular accepted set, split manifest, and schema versions. New accepted cases require a new release even if old files remain byte-identical.

4 Canonical HDF5 case schema

4.1 Why HDF5 is selected

HDF5 provides named multidimensional datasets, groups, attributes, typed numeric storage, chunking, and lossless compression. Its data model is suitable for self-describing scientific arrays and is supported by Python and MATLAB [1]. The format does not supply scientific meaning automatically; this module supplies the naming and semantic rules.

4.2 Schema identity

Every canonical case has root attributes

$$\text{schema_name} = \text{pinn_m3dc1.case}, \quad (4.1)$$

$$\text{schema_version} = 1.0.0, \quad (4.2)$$

$$\text{source_kind} = \text{synthetic_reduced_mhd}, \quad (4.3)$$

$$\text{axis_order} = \text{t,x,y}. \quad (4.4)$$

The schema name answers “which contract?” and the version answers “which revision?” A generic attribute such as **version=1** is insufficient because solver, equations, schema, and dataset releases evolve independently.

4.3 Required hierarchy

```

/
coordinates/
  x          float64 [Nx]
  y          float64 [Ny]
  t          float64 [Nt]
fields/
  psi        float64 [Nt, Nx, Ny]
  U          float64 [Nt, Nx, Ny]
  J          float64 [Nt, Nx, Ny]
  omega      float64 [Nt, Nx, Ny]
parameters/
  eta        float64 scalar
  nu         float64 scalar
  A0         float64 scalar
  psi_eq     float64 scalar
diagnostics/
  time_series/ float64 [Nt] datasets

```

events/	event value/index/presence datasets
physics/	versions, IC, BC, gauge, normalization
numerics/	grid, FFT, dealiasing, integrator, precision
verification/	acceptance flags, tolerances, report links
provenance/	source, config, platform, dependencies

The tree is normative at the group and dataset level. Implementations may add names only under a documented extension namespace or in a backward-compatible minor schema revision.

4.4 Dataset or attribute?

Use datasets for arrays, values that must be sliced, and values whose exact numeric dtype matters. Use attributes for small descriptive metadata attached to a group or dataset. Required scientific scalars such as η and ν are datasets rather than relying only on attributes, because scalar datasets have unambiguous paths, dtypes, and cross-language read behavior.

Large JSON strings, logs, and full configuration files should not be copied indiscriminately into attributes. Store compact identifiers and checksums inside the case, and preserve the authoritative text artifact beside the dataset release.

4.5 Coordinates

The coordinate datasets satisfy:

$$\text{shape}(x) = (N_x), \quad (4.5)$$

$$\text{shape}(y) = (N_y), \quad (4.6)$$

$$\text{shape}(t) = (N_t), \quad (4.7)$$

with strictly increasing values. Validators require

$$x_0 = 0, \quad 0 \leq x_i < L_x, \quad (4.8)$$

$$y_0 = 0, \quad 0 \leq y_j < L_y, \quad (4.9)$$

$$t_0 = 0, \quad t_{n+1} > t_n. \quad (4.10)$$

Uniform spacing is recorded and checked for the version-1 pseudo-spectral source. The explicit arrays remain authoritative even if `dx`, `dy`, and `save_interval` are also stored.

4.6 Canonical field axis order

All scalar fields use the logical order

$$q[n, i, j] = q(t_n, x_i, y_j), \quad q \in \{\psi, U, J, \omega\}. \quad (4.11)$$

Thus the stored shape is $[N_t, N_x, N_y]$. The axis names are stored with every field, and a reader validates dimensions against the three coordinate lengths.

Validation checkpoint : non-square sentinel

Axis parity cannot be verified reliably on a square grid containing symmetric fields. Every reader/writer test must include a non-square, non-symmetric sentinel array such as $q[n, i, j] = 10^4 n + 10^2 i + j$. A transpose then produces an unmistakable failure.

4.7 Memory layout is not logical axis order

NumPy commonly uses row-major contiguous arrays; MATLAB uses column-major storage. HDF5 defines dataset dimensions independently of either language's in-memory stride order. A language binding may require an explicit permutation or transpose to produce the canonical logical array. The reader contract is Equation (4.11), not “whatever shape the library returned.”

4.8 Field-specific attributes

Each field records:

- `long_name`;
- `axis_names = [t,x,y]`;
- `normalization = dimensionless_v1`;
- `units = 1` for dimensionless quantities;
- `definition_version`;
- `role = prognostic, reconstructed, or derived`; and
- the formula dependency for J and ω .

The string `units=1` means dimensionless. It does not mean unknown units.

4.9 Precision, special values, and endianness

Canonical reference fields, coordinates, parameters, and diagnostics are IEEE 754 binary64 values. All required numeric datasets must be finite. NaN and infinity are rejection conditions, not missing-value markers. Optional absent values use an explicit presence flag and a documented sentinel, never an unexplained NaN.

Writers use a declared standard little-endian float64 file datatype for portability. Readers convert to the native representation while preserving numerical values. Any later float32 learning cache is derived and must quantify its error relative to the float64 parent.

4.10 Chunking and lossless compression

Chunk shape affects performance but not scientific meaning. The default field chunk is

$$(1, \min(N_x, 64), \min(N_y, 64)), \quad (4.12)$$

which supports snapshot and spatial-block access without loading an entire trajectory. Lossless compression and byte shuffling may be used and must be recorded. A compression setting never changes the logical checksum defined for uncompressed values.

4.11 Diagnostics and event records

Time-aligned scalar diagnostics use length N_t . Required initial diagnostics are:

- magnetic energy;
- kinetic energy;
- total energy;
- Ohmic dissipation;
- viscous dissipation;
- maximum absolute current;
- reconnected flux; and
- reconnection rate when defined.

An event record stores at least `present`, `time`, `snapshot_index`, `value`, and `definition_version`. Events include current-sheet onset, first current peak, first reconnection-rate peak, topology merger, and confirmation of a post-peak interval. Interpolated event times must say which interpolation method was used.

5 Parameters, physics metadata, and normalization

5.1 Case parameters

The version-1 parameter vector is

$$\boldsymbol{\mu} = (\eta, \nu, A_0). \quad (5.1)$$

The equilibrium amplitude Ψ_{eq} is stored even when fixed at unity because it enters the initial-condition definition. Derived dimensionless groups may be stored for convenience,

$$S = 1/\eta, \quad \text{Re} = 1/\nu, \quad \text{Pm} = \nu/\eta, \quad (5.2)$$

but the primitive parameter datasets remain authoritative.

5.2 Physics metadata

The `/physics` group records:

- equations name and version;
- sign-convention version;
- dimensionality and coordinate system;
- domain periods and boundary conditions;
- guide-field assumption;
- density and incompressibility assumptions;
- initial-condition name, formula version, and parameters;
- gauge and mean policies;
- physical-normalization name and reference scales; and
- the declared relationship of stored primary and derived fields.

5.3 Two normalization layers

Let q_{dim} be a dimensional physical variable and q_0 a physical reference scale. Physical nondimensionalization gives

$$q = \frac{q_{\text{dim}}}{q_0}. \quad (5.3)$$

A later neural standardization may give

$$\tilde{q} = \frac{q - m_q}{s_q}. \quad (5.4)$$

The first transformation defines the PDE coefficients and physical interpretation. The second improves optimization conditioning. Equation (5.4) must be invertible:

$$q = m_q + s_q \tilde{q}, \quad q_{\text{dim}} = q_0(m_q + s_q \tilde{q}). \quad (5.5)$$

Important distinction

Calling both operations “normalization” without a namespace is dangerous. The contract uses `physical_normalization` for the reduced-MHD scale system and `preprocessing` for neural transforms.

5.4 Reference scales

The physical-normalization record contains the declared length, magnetic-field, density, velocity, time, flux, stream-function, current, and vorticity scales. If a synthetic study chooses numerical scale values of one, the symbolic definitions and dimensional units are still recorded. This preserves the chain needed for later comparison with dimensional exports.

6 Numerical metadata and verification evidence

6.1 Numerical realization metadata

The `/numerics` group records:

- spatial method and version;
- $N_x, N_y, \Delta x, \Delta y$;
- Fourier wave-number ordering and transform normalization;
- dealiasing rule;
- state variables actually advanced;
- Poisson zero-mode policy;
- time integrator and version;
- nominal and actual time-step policy;

- requested output times and actual saved times;
- floating-point precision; and
- restart or continuation ancestry.

The coordinate arrays and field arrays are authoritative values. Duplicated scalars such as N_x exist to make validation and search convenient; they must agree with array shapes.

6.2 Acceptance record

The `/verification` group stores one machine-readable flag per required test, the tolerance, measured value, pass/fail result, and test-definition version. A single `accepted=true` flag is not sufficient by itself. The aggregate status is recomputed from required component flags.

6.3 Convergence references

A production case can refer to a convergence campaign rather than embedding all refinement arrays. The case records the convergence-report identifier, dataset release, compared resolutions, observed differences, and the criterion that selected this realization. The referenced report is content-addressed and retained with the project artifacts.

6.4 Rejected and quarantined data

The allowed lifecycle states are:

$$\text{incomplete, candidate, accepted, rejected, quarantined.} \quad (6.1)$$

Only `accepted` can enter a release. `rejected` means a known criterion failed. `quarantined` means integrity or semantic compatibility is uncertain. State changes are recorded in provenance rather than erasing history.

7 Provenance, integrity, and reproducibility

7.1 What provenance must answer

For each realization, another researcher should be able to answer:

1. Which configuration initiated the run?
2. Which source revision and implementation produced it?

3. Which dependency and platform versions were used?
4. Which random inputs or perturbations were present?
5. Which verification results accepted it?
6. Has the file changed since the release was frozen?
7. Is this file a restart, migration, conversion, or derived product?

7.2 Required provenance fields

Store the UTC creation time, producer language, producer package version, source revision, dirty-worktree flag, configuration path and SHA-256 digest, command or entry-point identity, dependency lock digest when available, operating system, architecture, and parent artifact identifiers.

Do not store secrets, access tokens, full environment dumps, or private absolute paths. Reproducibility metadata must be useful without leaking credentials or unrelated user information.

7.3 Configuration preservation

The exact canonical run configuration is retained as a release artifact. The case stores its digest and the subset of values required for self-description. A default applied silently by code is still an input and must appear in the resolved configuration.

7.4 Checksums without self-reference

The dataset manifest stores a SHA-256 checksum of every closed HDF5 file. The checksum is external because a file cannot contain a checksum of its own final bytes without a self-reference problem. The case may also store logical payload digests computed over a declared canonical sequence of dataset paths, dtypes, shapes, and uncompressed values.

JSON manifests use a deterministic canonical serialization before hashing. A canonical JSON representation makes equivalent metadata produce the same byte sequence; RFC 8785 defines one such scheme [3].

7.5 Atomic publication

Cases are written under a temporary name, closed, reopened, schema-validated, physics-validated, and checksummed before atomic publication. The release manifest is written only after every referenced file exists and passes validation. An interrupted writer must not leave a partial file with an accepted production name.

Required result

Reproducibility requires both semantic identity and byte integrity. A matching checksum proves that bytes have not changed; it does not prove that the field sign, axis order, or split design is scientifically correct. Schema and physics validators provide that second layer.

8 Dataset release manifest

8.1 Why a second file is required

A case file describes one realization. A dataset manifest describes relationships across cases: membership, order, split assignment, intended task, checksums, duplicate groups, preprocessing, and release history. These facts cannot be represented reliably by inspecting filenames.

The version-1 manifest is UTF-8 JSON conforming to RFC 8259 [2]. JSON is selected because both Python and MATLAB can read it without depending on a language-specific object format. Numeric field arrays remain in HDF5; JSON is not used for bulk arrays.

8.2 Manifest identity

The top-level required fields are:

- `schema_name = pinn_m3dc1.dataset_manifest;`
- `schema_version = 1.0.0;`
- `dataset_id` and `release_version;`
- `created_utc;`
- `case_schema_requirement;`
- `physics_compatibility;`
- `ordered cases;`
- `split_manifest;`
- `preprocessing;`
- `integrity;` and
- `provenance.`

8.3 Case index entries

Each case entry stores:

- relative file path;
- file SHA-256 and byte size;
- `physical_case_id` and `case_id`;
- source kind and acceptance status;
- parameter vector;
- coordinate and field shapes;
- schema and conventions versions;
- split and test-regime labels; and
- any duplicate, refinement-family, or parity-family identifier.

Paths are relative to the manifest directory and use forward slashes. Moving a release directory therefore does not invalidate its internal path references.

8.4 One authoritative split assignment

The split may be represented as explicit lists of physical-case identifiers or as labels on case entries, but the manifest must have one authoritative form. If both are stored for convenience, validation requires exact agreement.

8.5 Release immutability

Once published, a release is immutable. Correcting a case, changing a split, adding a required field, or recomputing preprocessing creates a new release. An older release may be marked superseded, but its files and manifest remain available for reproduction.

This follows the central release principle of semantic versioning: published contents are not silently modified, and compatibility is communicated by major, minor, and patch components [4]. The schema and dataset release have independent versions because a dataset can add cases without changing its case schema.

9 Whole-case split construction

9.1 The scientific unit of generalization

The main surrogate question is whether a model predicts an unseen parameter combination. Therefore the split unit is the physical case, not an array element. Define disjoint sets

$$\mathcal{P}_{\text{tr}}, \quad \mathcal{P}_{\text{va}}, \quad \mathcal{P}_{\text{ti}}, \quad \mathcal{P}_{\text{te}}, \quad (9.1)$$

for training, validation, interpolation test, and extrapolation test physical-case identifiers. They satisfy pairwise disjointness and

$$\mathcal{P}_{\text{all}} = \mathcal{P}_{\text{tr}} \cup \mathcal{P}_{\text{va}} \cup \mathcal{P}_{\text{ti}} \cup \mathcal{P}_{\text{te}}. \quad (9.2)$$

All realizations, snapshots, and grid points sharing one physical-case identifier remain in the same partition.

9.2 Roles of the partitions

Partition	Permitted use	Forbidden use
Training	Update model weights; fit preprocessing; design training samplers	Final unbiased scoring
Validation	Select architecture, loss weights, schedules, stopping, and checkpoints	Updating ordinary model weights as labeled training data
Interpolation test	Final scoring on unseen parameter combinations supported by the training parameter region	Any design, scaling, or stopping decision
Extrapolation test	Final scoring outside the declared training support	Any design, scaling, or stopping decision

9.3 Interpolation in several parameters

For training parameter vectors $\{\mu_c : c \in \mathcal{P}_{\text{tr}}\}$, a held-out vector inside their convex hull is classified as parameter interpolation for the primary geometric rule. A vector outside that hull is extrapolation. Axis-aligned range labels are also stored because they are easy to interpret:

$$\eta_{\min}^{\text{tr}} \leq \eta \leq \eta_{\max}^{\text{tr}}, \quad \nu_{\min}^{\text{tr}} \leq \nu \leq \nu_{\max}^{\text{tr}}, \quad A_{0,\min}^{\text{tr}} \leq A_0 \leq A_{0,\max}^{\text{tr}}. \quad (9.3)$$

A point can satisfy all three coordinate ranges yet lie outside the convex hull. Both labels are useful and neither should be hidden inside a single combined test score.

9.4 Designed split before random split

When the parameter sweep is a structured grid, deliberately reserve interior combinations for interpolation and edge or outside-range combinations for extrapolation. Random splitting may accidentally create an easy or scientifically uninformative test set. Randomness is permitted only within a predeclared stratification rule and with a saved seed and algorithm version.

9.5 Duplicate and near-duplicate grouping

Before splitting, group cases that share:

- the same physics-defining parameters;
- the same initial condition under a different numerical resolution;
- Python and MATLAB parity realizations;
- restart-derived duplicates;
- migrated copies of an older schema; or
- numerically equivalent parameters within the declared canonicalization tolerance.

The group, not the file, is assigned to a partition.

9.6 No test information in preprocessing

After the split is frozen, every fitted statistic has the form

$$\hat{\alpha} = \text{FitTransform}(\mathcal{D}_{\text{tr}}), \quad (9.4)$$

where α contains means, scales, ranges, logarithmic-transform choices, clipping policies, or sampling weights. Validation and test cases are transformed using $\hat{\alpha}$ without refitting.

Important distinction : leakage through convenience

Computing a separate output mean or scale for each test trajectory reveals information about that trajectory. Even though no target value is placed directly into a training batch, the preprocessing has used the complete held-out field.

9.7 Split freezing and test access

The split manifest is written before model development. Test case identifiers can be visible for bookkeeping, but test values and metrics must not guide architecture, loss weights, sampling, training duration, or checkpoint selection. If repeated test inspection occurs, the set has become a validation set and a new untouched test release is required.

10 Learning-ready views and preprocessing

10.1 A view, not a replacement

A learning-ready view selects points and applies transforms to a canonical release. It records:

$$\text{view} = (\text{parent release}, \text{split}, \text{selection}, \text{transforms}, \text{output layout}). \quad (10.1)$$

The canonical HDF5 cases remain unchanged. Views can be regenerated and compared because their instructions and random seeds are explicit.

10.2 Raw neural map

The physical input and primary target are

$$\mathbf{s} = (x, y, t, \eta, \nu, A_0), \quad (10.2)$$

$$\mathbf{q} = (\psi, U). \quad (10.3)$$

The learned map is

$$(x, y, t, \eta, \nu, A_0) \mapsto (\hat{\psi}, \hat{U}). \quad (10.4)$$

J and ω remain evaluation fields and derivative-consistency references; they are not primary target channels in the frozen Module 4/5 comparison.

10.3 Periodic coordinate representation

Define

$$\phi_x = \frac{2\pi x}{L_x}, \quad \phi_y = \frac{2\pi y}{L_y}. \quad (10.5)$$

The primary spatial encoding is

$$\mathbf{e}_{\text{per}} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y). \quad (10.6)$$

The encoding is a model input transform, not a change to the stored coordinates. Raw (x, y) remain available for quadrature, plotting, residual interpretation, and inverse audit.

10.4 Time and amplitude scaling

For a declared training time interval,

$$\tilde{t} = 2 \frac{t - t_{\min}^{\text{tr}}}{t_{\max}^{\text{tr}} - t_{\min}^{\text{tr}}} - 1. \quad (10.7)$$

A_0 may use the same affine map based on training-case bounds or training statistics. Values outside $[-1, 1]$ are permitted at extrapolation time and must not be silently clipped.

10.5 Transport-coefficient transforms

If positive η and ν span several orders of magnitude, the declared transform may begin with

$$r_\eta = \log_{10} \eta, \quad r_\nu = \log_{10} \nu, \quad (10.8)$$

followed by training-only affine scaling. If the design is narrow and linear, raw coefficients may be scaled instead. The selected transform is part of the view version. An ideal case with zero coefficient cannot pass through a logarithm and requires an explicit separate policy.

10.6 Output standardization

The primary neural view stores

$$\tilde{\psi} = \frac{\psi - m_\psi}{s_\psi}, \quad \tilde{U} = \frac{U - m_U}{s_U}, \quad (10.9)$$

with m_ψ, s_ψ, m_U, s_U fitted from the declared training observations only. Required conditions are

$$s_\psi > 0, \quad s_U > 0. \quad (10.10)$$

The scientific prediction is recovered before evaluation:

$$\hat{\psi} = m_\psi + s_\psi \hat{\tilde{\psi}}, \quad \hat{U} = m_U + s_U \hat{\tilde{U}}. \quad (10.11)$$

10.7 Statistic weighting policy

A global pointwise mean weights cases in proportion to their number of stored points. If all accepted cases share the same grid and output count, this is equivalent to equal case weighting. If they differ,

the contract must choose explicitly between:

$$m_q^{\text{point}} = \frac{\sum_c \sum_{n,i,j} q_{nij}^{(c)}}{\sum_c N_t^{(c)} N_x^{(c)} N_y^{(c)}}, \quad (10.12)$$

$$m_q^{\text{case}} = \frac{1}{N_{\text{case}}} \sum_c \left[\frac{1}{N_t^{(c)} N_x^{(c)} N_y^{(c)}} \sum_{n,i,j} q_{nij}^{(c)} \right]. \quad (10.13)$$

The primary version-1 view uses case-balanced statistics so that a higher-resolution realization does not dominate merely by containing more entries.

10.8 Chain-rule metadata for the PINN

Module 5 differentiates inverse-transformed fields with respect to raw physical coordinates. Therefore the preprocessing record must expose analytic Jacobian factors. For example, if

$$\tilde{x} = (x - m_x)/s_x, \quad \psi = m_\psi + s_\psi \tilde{\psi}, \quad (10.14)$$

then

$$\partial_x \psi = \frac{s_\psi}{s_x} \partial_x \tilde{\psi}, \quad (10.15)$$

$$\partial_x^2 \psi = \frac{s_\psi}{s_x^2} \partial_x^2 \tilde{\psi}. \quad (10.16)$$

With sine-cosine encoding, automatic differentiation must traverse the phase map $2\pi x/L_x$. A black-box transform that cannot report these derivatives is not acceptable for the strong-form PINN.

10.9 Clipping and imputation

The reference fields contain no missing required values. The primary view performs no output clipping and no imputation. Clipping a test input to the training range changes extrapolation into a boundary prediction and must not occur silently. Any robustness experiment that adds noise, masks values, or clips outliers is a separately versioned view.

11 Flattening, sampling, and reconstruction

11.1 From arrays to supervised rows

For case c , the canonical array element gives

$$\mathbf{s}^{(c,n,i,j)} = (x_i, y_j, t_n, \eta_c, \nu_c, A_{0,c}), \quad (11.1)$$

$$\mathbf{q}^{(c,n,i,j)} = (\psi_{nij}^{(c)}, U_{nij}^{(c)}). \quad (11.2)$$

A flattened table is permitted as a cache, but every row retains:

$$(\text{case_id}, n, i, j, \text{split}, \text{selection_id}). \quad (11.3)$$

Those indices make the operation reversible.

11.2 Canonical linear index

For zero-based indices and axis order $[t, x, y]$, define

$$\ell = (nN_x + i)N_y + j. \quad (11.4)$$

Then

$$j = \ell \bmod N_y, \quad (11.5)$$

$$i = \left\lfloor \frac{\ell}{N_y} \right\rfloor \bmod N_x, \quad (11.6)$$

$$n = \left\lfloor \frac{\ell}{N_x N_y} \right\rfloor. \quad (11.7)$$

This is a logical enumeration for interchange. A language may use a different native linear-memory order internally, but serialized observation indices use Equation (11.4).

11.3 Observation sets

An observation set is a deterministic selection of rows for one data budget. It stores:

- parent dataset and split manifest digests;
- per-case selected (n, i, j) indices;
- selection algorithm and version;
- random seed;
- desired and realized counts;
- replacement policy;
- stratification by time or structure; and
- its own checksum.

Module 4 and Module 5 use the same observation-set identifier in a fair comparison.

11.4 Hierarchical sampling

The recommended training sampler chooses a case, then a time, then spatial points. This makes it possible to balance cases and prevent dense snapshots from controlling the loss. The exact probabilities are saved. Uniform point sampling over all concatenated arrays is a different distribution and must be labeled as such.

11.5 Ragged cases

Cases may eventually have different N_x, N_y, N_t or saved times. The canonical release stores each case independently and does not pad reference fields. A learning view may concatenate rows because coordinate networks accept variable grids. Any padded tensor representation must store masks and may not treat padding as a physical zero.

11.6 Full-grid evaluation remains authoritative

Training observations are a subset. Validation and test scoring reconstruct full declared evaluation grids and compare all four fields and diagnostics. A good error on the sampled training rows is not evidence of correct topology, derivative fields, or current-sheet structure.

12 Schema and physics validation

12.1 Validation layers

Layer	Examples	Failure meaning
Structural	Required paths, dtypes, ranks, shapes, strings	File does not implement the schema
Relational	Field dimensions match coordinates; indices are valid	Internally inconsistent serialization
Numerical	Finite values, monotone coordinates, gauge tolerance	Invalid numerical contents
Physical	$J = -\nabla_{\perp}^2 \psi$, $\omega = \nabla_{\perp}^2 U$, periodicity, diagnostics	Stored meanings or computations disagree
Provenance	Required versions, config digest, parent links	Reproduction chain is incomplete
Release	Checksums, membership, disjoint splits, no duplicates	Dataset assembly is corrupted or leaky

12.2 Shape and coordinate checks

For every field q ,

$$\text{shape}(q) = (\text{len}(t), \text{len}(x), \text{len}(y)). \quad (12.1)$$

The validator checks endpoint exclusion, periodic grid spacing, time monotonicity, declared domain lengths, and agreement between redundant metadata and authoritative arrays.

12.3 Derived-field consistency

On the periodic grid, recompute

$$J_{\text{check}} = -\nabla_{\perp,h}^2 \psi, \quad \omega_{\text{check}} = \nabla_{\perp,h}^2 U \quad (12.2)$$

with the declared discrete spectral operator. Report both absolute and normalized errors, for example

$$\epsilon_J = \frac{\|J - J_{\text{check}}\|_2}{\max(\|J\|_2, \epsilon_{\text{floor}})}. \quad (12.3)$$

The tolerance is versioned and must reflect roundoff, output precision, and whether stored fields were computed before or after any spectral projection.

12.4 Gauge and mean checks

For every snapshot,

$$|\langle U \rangle_h| \leq \tau_U. \quad (12.4)$$

The mean of ψ is compared with its recorded initial value. A case-dependent constant offset cannot be repaired silently during ingestion; repair would create a migrated realization with explicit ancestry.

12.5 Diagnostic recomputation

Recompute energies, maximum current, and selected topology diagnostics from stored fields. Stored histories must agree within declared tolerances. This detects stale diagnostics written from an earlier in-memory state or an inconsistent axis permutation.

12.6 Manifest checks

The release validator requires:

- every indexed file exists and matches its checksum;

- every case is accepted and schema-compatible;
- no unindexed file is treated as release data;
- physical-case groups are split exactly once;
- train, validation, and test groups are disjoint;
- preprocessing depends only on training identifiers;
- observation indices lie inside their parent arrays; and
- all parent digests and version constraints resolve.

Validation checkpoint : fail before concatenation

Compatibility is checked case by case before any arrays are concatenated. Once incompatible arrays have been flattened together, their original meanings can be difficult or impossible to recover.

13 Python–MATLAB interoperability

13.1 Scientific parity, not identical internal code

Python and MATLAB readers and writers may use different APIs and internal layouts. They must nevertheless agree on the logical HDF5 paths, axis names, numeric values, string encoding, identifiers, checksums, and validation outcomes.

13.2 Mandatory round-trip matrix

Writer	Reader	Required result
Python	Python	Exact schema and value round trip
Python	MATLAB	Canonical logical arrays after explicit mapping
MATLAB	MATLAB	Exact schema and value round trip
MATLAB	Python	Canonical logical arrays after explicit mapping

Every direction uses a small non-square sentinel case and at least one production-like case.

13.3 Strings, booleans, and scalars

Cross-language failures often arise from metadata rather than fields. The contract therefore specifies UTF-8 strings, scalar numeric datasets with rank zero, and acceptance flags stored as unsigned 8-bit

integers with allowed values 0 and 1. Readers convert these representations to native strings and logical values only after schema validation.

Fixed versus variable-length strings must be tested by both readers. Trailing NUL bytes or padded spaces are not part of an identifier. The canonical comparison uses decoded Unicode text.

13.4 Index bases

Serialized snapshot and grid indices are zero based. MATLAB converts to one-based subscripts at its internal boundary:

$$n_{\text{MATLAB}} = n_{\text{stored}} + 1, \quad (13.1)$$

and similarly for i and j . Writing a MATLAB index without subtracting one creates a valid-looking but shifted observation set.

13.5 Numeric comparison

Values written and read without arithmetic should agree bitwise after dtype conversion. Independently generated solver results are compared with declared physical tolerances, not required to be bitwise identical. The report must distinguish an I/O round-trip test from a solver-parity test.

14 Meaning of M3D-C1-shaped I/O

14.1 The similarity being claimed

The synthetic workflow and a future M3D-C1 workflow can share an abstract record:

$$\text{source metadata} + \text{coordinates/mesh} + \text{parameters} + \text{time-dependent fields} + \text{diagnostics} + \text{splits}. \quad (14.1)$$

This is the sense in which the I/O is M3D-C1-shaped. It demonstrates a disciplined route by which high-fidelity exports could become learning data.

14.2 What cannot be identified by name alone

Synthetic version 1	Possible M3D-C1 export concept	Required adapter question
Cartesian (x, y) periodic grid	Finite-element mesh in toroidal geometry	Which coordinates, elements, basis functions, periodic identifications, and quadrature define the field?

Synthetic version 1	Possible M3D-C1 export concept	Required adapter question
ψ Cartesian flux function	Flux-like exported potential	Are sign, gauge, dimensional scale, geometric factors, and physical definition equivalent?
U Cartesian stream function	Flow-like reduced potential	Which velocity components and metric factors does it generate?
$J = -\nabla_{\perp}^2 \psi$	Current component or current density	Which component, operator, units, sign, and basis are used?
$\omega = \nabla_{\perp}^2 U$	Vorticity-like variable	Is it exported, reconstructible, and defined by the same operator?
η, ν, A_0	Transport, equilibrium, and perturbation inputs	Are coefficients scalar, spatially varying, tensorial, dimensional, or closure-dependent?
Two-field incompressible model	Selected extended/reduced MHD option	Which evolved fields, closures, sources, and equations were active?

M3D-C1 adapter boundary

Matching symbols do not establish matching variables. The adapter must derive or document the map from exported quantities to canonical learning variables, including geometry, units, normalization, sign, gauge, interpolation, and information loss.

14.3 Source-native and canonical layers

Genuine exports first enter an immutable **source-native** layer that preserves the original mesh, field names, units, and metadata. An adapter produces a separate canonical layer. It never overwrites the source. The canonical artifact records:

- source file checksums and export-tool version;
- selected M3D-C1 model option and equations;
- field-by-field mapping formulas;
- coordinate and mesh transformation;
- interpolation or projection operator;
- unit and normalization conversion;
- sign and gauge corrections;

- fields dropped, approximated, or synthesized;
- conservation and reconstruction errors; and
- adapter validation results.

14.4 Structured grid is not a universal canonical target

Forcing finite-element data immediately onto the synthetic $[t, x, y]$ grid may erase mesh geometry, basis order, curved elements, and metric information. The future schema therefore uses a source-family discriminator. `synthetic_reduced_mhd` requires the structured-grid groups defined in Section 4; `m3dc1_native` will require a separate major source schema with mesh topology and field association. A later learning view may project both onto a common representation only after projection error is measured.

14.5 Conditions for an M3D-C1 surrogate claim

The project may use the phrase “validated M3D-C1 surrogate” only after:

1. genuine exports are available across a declared M3D-C1 configuration domain;
2. active equations, closures, geometry, boundary conditions, mesh, and units are known;
3. the adapter passes field and diagnostic reconstruction tests;
4. train/validation/test splits contain independent M3D-C1 cases;
5. accuracy is evaluated against untouched genuine outputs; and
6. discrepancies between the synthetic reduced model and M3D-C1 are reported rather than hidden.

15 Versioning, compatibility, and migration

15.1 Independent version namespaces

The following versions are stored separately:

- case schema;
- dataset-manifest schema;
- equations and sign conventions;

- physical normalization;
- initial-condition formula;
- diagnostic definitions;
- solver implementation;
- preprocessing view;
- split manifest; and
- dataset release.

Changing one does not imply that all others changed.

15.2 Semantic version policy

For a public schema $X.Y.Z$:

- increment Z for a correction that does not change valid instance interpretation;
- increment Y for backward-compatible optional additions; and
- increment X for incompatible names, required fields, shapes, dtypes, axes, signs, units, or meanings.

A change from $[t, x, y]$ to $[t, y, x]$, from $J = -\nabla_{\perp}^2 \psi$ to $J = +\nabla_{\perp}^2 \psi$, or from dimensionless to dimensional values is a major change even if a converter is easy to write.

15.3 Reader behavior

A reader declares a version range it supports. It rejects an unknown major version. It may accept a newer minor version only if required names retain their meanings and unknown optional extensions can be ignored safely. It never guesses a missing version from a filename or date.

15.4 Migration rules

A migration is a deterministic, versioned transformation

$$\mathcal{M}_{a \rightarrow b} : D_a \longrightarrow D_b. \quad (15.1)$$

It produces a new file with a new `case_id`, preserves the old file, records parent checksums, writes the migration tool revision, and validates both generic structure and known scientific invariants.

Lossless renaming differs from scientific conversion. If a migration changes precision, interpolates a grid, changes a gauge, or reconstructs a field, the numerical error and information loss are reported.

15.5 Deprecation

Deprecated optional names may be read for a stated transition period but are never written by the newest writer. Required names cannot simply disappear in a minor revision. The migration guide lists replacement paths and any change in meaning.

16 Common failure modes

16.1 A transposed field that still looks plausible

On a square symmetric benchmark, exchanging x and y can preserve the shape and produce visually reasonable contours. Non-square sentinels, asymmetric analytic values, and coordinate-aware diagnostics are necessary.

16.2 Endpoint duplication

Storing both $x = 0$ and $x = L_x$ duplicates a periodic point. It changes quadrature weights, creates repeated training rows, and may produce an apparent seam if the values differ by roundoff or interpolation.

16.3 Silent float32 conversion

Single precision may preserve broad ψ contours while degrading J , ω , high-order PINN residuals, and small balance defects. A float32 cache is an evaluated derivative product, not the reference dataset.

16.4 Stale derived fields

A writer can update ψ but accidentally store J from an earlier state. Recomputing J and ω on read detects this.

16.5 One accepted flag with no evidence

A Boolean cannot explain which tolerances were applied or which test failed after a definition changed. Store the component results and recompute the aggregate.

16.6 Random point split

Neighboring points from the same trajectory appear in both train and test. The resulting score measures interpolation within seen physical cases and overstates parameter generalization.

16.7 Normalization before splitting

Global means, standard deviations, parameter ranges, or clipping thresholds incorporate held-out cases. The test distribution has influenced the training pipeline even if its rows never enter a gradient batch.

16.8 Filename as metadata

Parsing parameters from filenames loses type, unit, formula version, and canonical precision. Rename operations can silently change apparent meaning. Filenames assist humans; dataset contents and manifests are authoritative.

16.9 Plots as training data

Images discard field precision, coordinates, gauge, units, sign, and the relationship among variables. They are visualization artifacts, not canonical targets.

16.10 Schema-valid but physics-invalid

A file can have all required paths and still store J with the wrong sign. Structural validation and physical validation are separate mandatory layers.

16.11 M3D-C1 name matching

Mapping a field named `psi` directly to the synthetic ψ without checking geometry, normalization, gauge, and definition is semantic corruption disguised as convenience.

17 Module 3 computational contract

17.1 Inputs

Module 3 receives:

- closed Module 2 candidate case files;

- complete verification records;
- resolved run configurations;
- equations, conventions, IC, diagnostic, and normalization versions; and
- a version-controlled dataset-release configuration.

17.2 Outputs

It produces:

- immutable accepted canonical HDF5 cases;
- rejected or quarantined cases outside the accepted index;
- a versioned JSON dataset manifest;
- a whole-physical-case split manifest;
- training-only preprocessing records;
- optional reproducible observation-set manifests;
- integrity checksums; and
- validation and Python–MATLAB round-trip reports.

17.3 Recommended components

Component	Responsibility
Case inspector	Read candidate metadata without accepting it implicitly.
Schema validator	Validate names, versions, dtypes, ranks, shapes, and allowed values.
Physics validator	Recompute identities, gauge, periodicity, and diagnostic checks.
Identity builder	Construct physical-case and realization identifiers from canonical metadata.
Canonical writer	Publish atomic HDF5 cases in schema version 1.0.0.
Manifest builder	Index membership, hashes, parameters, schemas, and ancestry.
Duplicate grouper	Keep equivalent physical cases and realizations in one split group.
Split builder	Create designed whole-case partitions and freeze their digest.
Preprocessor fitter	Fit invertible transforms on training observations only.

Component	Responsibility
Observation builder	Store deterministic per-case indices for controlled data budgets.
Round-trip suite	Test Python/MATLAB readers and writers with sentinel and production cases.
Migration tool	Convert an explicit old schema to a new one without overwriting parents.

17.4 Required test families

- missing path, wrong dtype, wrong rank, and wrong-axis negative tests;
- non-square sentinel round trips in all language directions;
- field identity, gauge, periodicity, and diagnostic recomputation;
- checksum corruption detection;
- duplicate physical-case grouping;
- split disjointness and coverage;
- test-leakage guards for preprocessing dependencies;
- flatten/unflatten exact recovery;
- observation-set reproducibility;
- schema-version acceptance and rejection matrix; and
- migration ancestry and scientific-invariant checks.

17.5 Completion criteria

Module 3 implementation is complete only when:

1. at least one accepted Module 2 trajectory is serialized and revalidated;
2. both languages read the same logical case values and metadata;
3. a release manifest passes integrity and membership checks;
4. all physical-case groups are assigned exactly once;
5. preprocessing proves training-only dependency;
6. full field reconstruction from observation indices passes;

7. deliberate corruption and leakage cases are rejected; and
8. a clean command reproduces the release from recorded configurations.

Important distinction

This physics note freezes the intended contract. The current repository module remains an implementation scaffold until the writers, validators, manifests, and tests above actually exist and pass.

18 Worked miniature example

18.1 A tiny accepted realization

Suppose a sentinel case has

$$N_t = 3, qquad N_x = 4, qquad N_y = 5. \quad (18.1)$$

Then every field has shape $[3, 4, 5]$ and contains 60 values. Define

$$q[n, i, j] = 10^4 n + 10^2 i + j. \quad (18.2)$$

The entry $q[2, 3, 4] = 20304$. Any exchange of x and y , one-based shift, or time-axis reversal changes this expected value.

18.2 Flatten and invert

For $(n, i, j) = (2, 3, 4)$, Equation (11.4) gives

$$\ell = (2 \cdot 4 + 3) \cdot 5 + 4 = 59. \quad (18.3)$$

Inversion gives

$$j = 59 \bmod 5 = 4, \quad (18.4)$$

$$i = \lfloor 59/5 \rfloor \bmod 4 = 3, \quad (18.5)$$

$$n = \lfloor 59/20 \rfloor = 2. \quad (18.6)$$

Thus the logical last element is recovered exactly, independent of native language indexing.

18.3 Physical-case grouping example

Suppose three files share $\eta = 10^{-3}$, $\nu = 10^{-3}$, $A_0 = 0.1$ and the same physics definitions:

- Python 128² production realization;
- MATLAB 128² parity realization; and
- Python 256² refinement realization.

They have three `case_id` values but one `physical_case_id`. If the production realization is assigned to training, the other two cannot be used as unseen test cases. They may support training-only verification or be excluded from learning releases.

18.4 Training-only scale example

Assume the training observations have

$$m_\psi = 0.02, qquads_\psi = 0.71. \quad (18.7)$$

A held-out value $\psi = 0.80$ becomes

$$\tilde{\psi} = \frac{0.80 - 0.02}{0.71} \approx 1.099. \quad (18.8)$$

The fact that this transformed value exceeds one is not an error. Refitting the scale on the held-out case or clipping it would change the evaluation question.

18.5 Acceptance logic

If the HDF5 file is structurally valid but the recomputed current gives

$$\frac{\|J + \nabla_{\perp,h}^2 \psi\|_2}{\|J\|_2} = 2 \times 10^{-3} \quad (18.9)$$

against a tolerance of 10^{-10} , the case is rejected or quarantined. It is not repaired by renaming J or loosening the tolerance after seeing the result. The producer or convention mismatch must be diagnosed.

19 Summary and bridge to Module 4

Module 3 turns accepted numerical trajectories into a stable scientific interface. The canonical case preserves raw dimensionless coordinates, float64 fields, parameters, diagnostics, equations, numerical settings, verification evidence, and provenance. Logical axes are explicitly $[t, x, y]$; $J = -\nabla_{\perp}^2 \psi$ and $\omega = \nabla_{\perp}^2 U$ remain checkable identities; the zero-mean gauge of U is enforced and audited.

The dataset manifest freezes release membership, file integrity, physical-case grouping, split assignment, and preprocessing. Complete physical cases, including all numerical realizations of the same initial-value problem, remain in one partition. Interpolation and extrapolation tests are separated. Every fitted transform uses training data only, remains invertible, and exposes the chain-rule information required by the PINN.

The phrase M3D-C1-shaped is intentionally limited. It means that the workflow anticipates mesh or coordinate metadata, parameters, evolving fields, diagnostics, provenance, and an explicit adapter. It does not identify the synthetic Cartesian potentials with code-native M3D-C1 variables. Genuine exports first remain source-native, then enter a versioned mapping whose geometry, units, signs, gauges, and projection errors are validated.

Module 4 receives this frozen release and constructs the fair data-only baseline. It may sample and standardize the accepted fields, but it cannot change the split, refit scales on held-out cases, erase case identity, or reinterpret the stored physics. Module 5 receives the identical labeled observations and adds physics constraints. That controlled inheritance is the reason Module 3 is part of the scientific method rather than merely storage engineering.

A Normative schema dictionary

Path or attribute	Type	Shape	Meaning
@schema_name	UTF-8	scalar	pinn_m3dc1.case
@schema_version	UTF-8	scalar	Case contract version
@source_kind	UTF-8	scalar	synthetic_reduced_mhd
@physical_case_id	UTF-8	scalar	Physics-equivalence group
@case_id	UTF-8	scalar	Unique numerical realization
@axis_order	UTF-8	scalar	Literal t,x,y
/coordinates/x	float64	$[N_x]$	Half-open periodic x grid
/coordinates/y	float64	$[N_y]$	Half-open periodic y grid
/coordinates/t	float64	$[N_t]$	Explicit saved times
/fields/psi	float64	$[N_t, N_x, N_y]$	Magnetic-flux function
/fields/U	float64	$[N_t, N_x, N_y]$	Zero-mean stream function
/fields/J	float64	$[N_t, N_x, N_y]$	$-\nabla_{\perp}^2 \psi$
/fields/omega	float64	$[N_t, N_x, N_y]$	$+\nabla_{\perp}^2 U$
/parameters/eta	float64	scalar	Magnetic diffusivity
/parameters/nu	float64	scalar	Kinematic viscosity
/parameters/A0	float64	scalar	Initial flow amplitude
/parameters/psi_eq	float64	scalar	Equilibrium flux amplitude
/diagnostics/time_series/*	float64	$[N_t]$	Time-aligned diagnostic
/diagnostics/events/*/present	uint8	scalar	0 or 1
/diagnostics/events/*/time	float64	scalar	Event time if present
/diagnostics/events/*/snapshot_index	int64	scalar	Zero-based nearest/indexed snapshot
/physics/*	mixed	scalar	Equations, IC, BC, gauge, normalization
/numerics/*	mixed	scalar	Grid, FFT, integrator, precision
/verification/*	mixed	scalar	Criteria, values, tolerances, flags
/provenance/*	mixed	scalar	Source, config, platform, ancestry

B Illustrative dataset manifest

The example below is explanatory. The machine implementation is validated against the separately versioned manifest schema.

```

{
  "schema_name": "pinn_m3dc1.dataset_manifest",
  "schema_version": "1.0.0",
  "dataset_id": "island-coalescence-synthetic-v1",
  "release_version": "1.0.0",
  "case_schema_requirement": ">=1.0.0,<2.0.0",
  "physics_compatibility": {
    "equations_version": "reduced_mhd_v1",
    "conventions_version": "conventions_v1",
    "physical_normalization": "dimensionless_v1"
  },
  "cases": [
    {
      "path": "cases/syn-r-54b3027976ad.h5",
      "sha256": "<64 lowercase hexadecimal characters>",
      "physical_case_id": "syn-p-8f3a10c64e9b",
      "case_id": "syn-r-54b3027976ad",
      "parameters": {"eta": 0.001, "nu": 0.001, "A0": 0.1},
      "split": "train",
      "acceptance_status": "accepted"
    }
  ],
  "split_manifest": {
    "unit": "physical_case_id",
    "algorithm": "designed_parameter_split_v1",
    "train": ["syn-p-8f3a10c64e9b"],
    "validation": [],
    "test_interpolation": [],
    "test_extrapolation": []
  },
  "preprocessing": {
    "fit_partition": "train",
    "view_version": "coordinate_mlp_v1",
    "record_path": "preprocessing/coordinate_mlp_v1.json"
  }
}

```

C Compatibility decision table

Change	Version effect	Reason
Correct prose without changing machine interpretation	Patch	No instance meaning changes
Add an optional diagnostic with a unique path	Minor	Old readers can ignore it safely
Add a new required dataset	Major	Old valid instances no longer satisfy the contract
Rename <code>psi</code> to <code>flux</code>	Major	Required path changes
Change axis order to $[t, y, x]$	Major	Every field index changes meaning
Change float64 reference fields to float32	Major	Precision contract changes
Change current sign	Major plus conventions version	Physical meaning changes
Change IC formula at the same parameters	IC version and new physical ID	Different initial-value problem
Add accepted cases without changing schema	Dataset release increment	Membership changes, schema does not
Move files while preserving relative manifest structure	New packaged release or identical logical release	Byte/path inventory must remain auditable

D Acceptance checklist

Per-case checks

- ☐ Required schema, source, case, physical-case, and axis attributes exist.
- ☐ Coordinate arrays are float64, finite, monotone, and consistent with the half-open domain.
- ☐ Every field is float64 with logical shape $[N_t, N_x, N_y]$.
- ☐ Parameters are finite and agree with the resolved configuration.
- ☐ $J = -\nabla_{\perp,h}^2 \psi$ and $\omega = \nabla_{\perp,h}^2 U$ pass declared tolerances.
- ☐ U satisfies its zero-mean gauge at every saved time.
- ☐ Diagnostics recompute within tolerance.
- ☐ All required Module 2 verification components pass.

- ☐ Provenance, source revision, config digest, and dependencies are present.
- ☐ File reopens cleanly after atomic publication.

Release checks

- ☐ Every manifest file exists, matches its checksum, and is accepted.
- ☐ No physical-case identifier appears in more than one partition.
- ☐ Every accepted release member has exactly one split assignment.
- ☐ Interpolation and extrapolation test labels are reproducible.
- ☐ Preprocessing dependencies are a subset of training case identifiers.
- ☐ Observation indices are valid and reproduce identical selections.
- ☐ Python and MATLAB readers agree on sentinel and production logical arrays.
- ☐ Dataset, split, preprocessing, and observation manifests have independent digests.
- ☐ Release documentation states the claim boundary and known limitations.

E Symbol table

Symbol	Meaning	Status
x, y	Periodic Cartesian coordinates	dimensionless
t	Saved physical time coordinate	dimensionless
N_x, N_y, N_t	Coordinate and snapshot counts	integers
L_x, L_y	Domain periods	2π baseline
ψ	Magnetic-flux function	stored float64
U	Flow stream function	stored float64
J	Current variable $-\nabla_{\perp}^2 \psi$	stored/checkable
ω	Vorticity $+\nabla_{\perp}^2 U$	stored/checkable
η	Magnetic diffusivity	case parameter
ν	Kinematic viscosity	case parameter
A_0	Initial flow perturbation amplitude	case parameter
Ψ_{eq}	Equilibrium flux amplitude	stored parameter
$\boldsymbol{\mu}$	Parameter vector (η, ν, A_0)	physical-case input
$\mathcal{D}$	Dataset release object	cases plus contracts
$\mathcal{P}_{\text{tr}}$	Training physical-case set	split
$\mathcal{P}_{\text{va}}$	Validation physical-case set	split
$\mathcal{P}_{\text{ti}}$	Interpolation-test physical-case set	split

Symbol	Meaning	Status
$\mathcal{P}_{\text{te}}$	Extrapolation-test physical-case set	split
m_q, s_q	Training-only location and scale for q	preprocessing
$\tilde{q}$	Neural transformed value	derived view
ℓ	Canonical zero-based flattened index	integer
τ_U	Gauge tolerance	versioned criterion

F Acronym and terminology table

Term	Meaning
API	Application programming interface
Canonical	The authoritative representation defined by the current contract
Case	One complete numerical realization unless qualified as physical case
Checksum	Digest used to detect changed bytes or logical payloads
Dataset	A complete release, or an HDF5 multidimensional array when used in HDF5 context
FAIR	Findable, Accessible, Interoperable, and Reusable data principles
Gauge	Convention fixing a physically irrelevant additive freedom
HDF5	Hierarchical Data Format version 5
I/O	Input/output
IC	Initial condition
JSON	JavaScript Object Notation, a language-independent text interchange format
Manifest	Authoritative index of release membership and relationships
M3D-C1	Extended-MHD simulation framework based on high-order C^1 finite elements
MHD	Magnetohydrodynamics
Migration	Versioned conversion from one schema to another
Normalization	Physical nondimensionalization; neural transforms use the separate term preprocessing
Observation set	Immutable selected indices used for one labeled-data budget
PINN	Physics-informed neural network
Provenance	Recorded production and ancestry history of an artifact
Realization	One numerical solution of a physical case under specified numerical settings
RMHD	Reduced magnetohydrodynamics; the exact reduction must be stated
Schema	Versioned rules for required names, types, shapes, and meanings

Term	Meaning
SHA-256	Secure Hash Algorithm with a 256-bit digest, used here for integrity
Split leakage	Held-out information entering training, design, or preprocessing
UTF-8	Unicode text encoding used by manifest and metadata strings
View	Reproducible derived representation for a learning task

G Concept questions and answer sketches

Questions

- Q1.** Why is a field name a physical statement?
- Q2.** Why are both `physical_case_id` and `case_id` required?
- Q3.** Why can a square grid hide an axis transposition?
- Q4.** Why are the canonical fields not standardized in place?
- Q5.** Why are J and ω stored if they can be derived?
- Q6.** What is the difference between physical nondimensionalization and neural preprocessing?
- Q7.** Why does the time array remain authoritative even with a recorded save interval?
- Q8.** Why is the file checksum stored outside the HDF5 file?
- Q9.** Why must missing acceptance metadata fail closed?
- Q10.** Why is a random point split not an unseen-case test?
- Q11.** Why do refinement and parity realizations belong to the same split group?
- Q12.** When can a held-out parameter vector be called interpolation?
- Q13.** Why are test-fitted means a form of leakage?
- Q14.** Why must observation indices retain case and grid identity?
- Q15.** Why is a padded zero not necessarily a physical zero?
- Q16.** What is the difference between I/O round-trip parity and solver parity?
- Q17.** Which schema changes require a major version increment?
- Q18.** Why does a matching checksum not prove correct physics?

Q19. What does M3D-C1-shaped mean in this project?

Q20. Why must a genuine M3D-C1 export remain source-native before adaptation?

Q21. What additional evidence is required for an M3D-C1 surrogate claim?

Answer sketches

- A1.** The name selects a definition, sign, normalization, gauge, and relationship to other fields; numbers alone do not.
- A2.** One groups scientifically equivalent problems for leakage control; the other identifies a particular numerical artifact.
- A3.** Exchanging equal-length axes preserves shape, and symmetry can preserve visual appearance.
- A4.** Standardization is training-release dependent and can be regenerated; overwriting destroys the reference and complicates physics derivatives.
- A5.** They permit derivative-sensitive evaluation, stale-field detection, and independent consistency audits.
- A6.** The first defines the dimensionless PDE; the second conditions numerical optimization and must be inverted.
- A7.** Adaptive or event-aligned output may not be uniform, and explicit coordinates prevent reconstruction ambiguity.
- A8.** A closed file cannot contain the digest of its own final bytes without changing those bytes.
- A9.** Eligibility requires affirmative evidence; absence cannot be interpreted as success.
- A10.** It places correlated points from the same parameter trajectory in both training and test.
- A11.** They solve the same physical initial-value problem and would otherwise create near-duplicate leakage.
- A12.** Under the primary rule, when it lies inside the convex hull of training parameter vectors; axis-wise labels supplement this.
- A13.** They use target-distribution information from a supposedly untouched case.
- A14.** Full fields, case-balanced losses, and exact audit require reversible mapping to parent arrays.
- A15.** Padding is a tensor-shape device and must be masked; zero may be a valid field value.
- A16.** Round-trip parity checks serialization; solver parity compares independently computed numerical solutions.

- A17.** Changes to required names, axes, dtypes, signs, units, shapes, or physical meanings.
- A18.** Integrity says bytes are unchanged, not that the original interpretation was correct.
- A19.** The workflow has an export-like field/coordinate/parameter/provenance interface while using synthetic reduced-MHD data.
- A20.** The original mesh, units, names, and metadata are needed to validate any later translation and measure information loss.
- A21.** Genuine independent exports, a validated adapter, known equations and closures, held-out M3D-C1 cases, and renewed scoring.

H References and standards

References

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